Xi Xie

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EDUCATION

- University of Connecticut — Storrs, CT, United States
  Ph.D. in Computer Science and Engineering; GPA: ***4.0****/4.0* — *Aug. 2022 - Present*
  Research Interests: ML Model/ CUDA Kernel Co-Design, Privacy-Preserving Machine Learning.
- Institute of Geophysics, China Earthquake Administration — Beijing, China
  Master's Degree in Geophysics — *Sep. 2015 - Oct. 2020*
- Beijing University of Technology — Beijing, China
  Bachelor's Degree in Software Engineering — *Sep. 2009 - Jul. 2013*

WORK AND RESEARCH EXPERIENCE

Linear Algebra Libraries Intern., NVIDIA, CA/CT-Remote, United States — May. 2025 – Jan.2026

- Designed and implemented production-level BSR-format SpMM GPU kernels using CuTe and cuBLASDx, achieving 1.87× / 2.03× average speedup over existing cuSPARSE implementations.
- Designed and implemented production-level BSR-format SpMV GPU kernels, achieving 2.35× average speedup over existing cuSPARSE implementations.
- Contributed to the cuSPARSE library within a large-scale CUDA C++ codebase, delivering generic API updates, kernel integration, performance benchmarking, and ongoing maintenance as part of the Math Libraries team using Git/Perforce workflows.
- Independently evaluated and integrated CuTe as a GPU programming abstraction newly adopted within the cuSPARSE library, enabling measurable performance gains while maintaining library-level correctness and compatibility.

Research Assistant, University of Connecticut, CT, United States — Aug. 2022 – May. 2025

- GPU Kernel and ML Model Co-Design for Performance Optimization.
  - Designed RTopK, an ultra-fast row-wise top-k kernel based on binary search, optimized for parallel top-k selection on large batches of limited-size vectors. This design achieves an average speedup of up to 11.49× with early stopping and 7.29× without early stopping compared to PyTorch.
  - Optimized GNN workflows by improving GPU kernel design, achieving state-of-the-art performance in GNN training and inference. The SpMM kernel design incorporated lightweight graph preprocessing, block-level partitioning, and a combined warp strategy, resulting in an average of 1.17× speedup over cuSPARSE. Also, developed a lightweight C++/ CUDA SpMM testing framework.
  - Designed MaxK-GNN, an innovative GNN training and inference acceleration framework introducing MaxK-nonlinearity. Developed novel SpMM kernel variants leveraging MaxK-nonlinearity to accelerate the SpMM operations, a key bottleneck in GNN workflows. These kernels achieved up to a 10.6× operator-level speedup and a 3.5× overall speedup with no accuracy degradation.
  - Earned 100+ stars on my three GitHub repositories for the above implementations:
    - github.com/xiexi51/RTopK
    - github.com/xiexi51/ICCAD-Accel-GCN
    - github.com/xiexi51/MaxK-GNN
  - Publications: 25'ICLR[1], 23'ICCAD[10], 24'ASPLOS[8], 25'ICS[5], 25'ICCAD[2]
- Fully/Partial Polynomial Model Design for Privacy-Preserving Computation Acceleration.
  - (Ongoing) Proposed PolyNorm, achieving state-of-the-art stable and accurate training for fully polynomial models on large-scale tasks; leveraged Distributed Data Parallel (DDP) for efficient training.

- Designed partial polynomial replacement methods for deep neural networks using a smoothing loss function to determine replacement patterns.
- Publications: 24'ICCAD[9], 23'ICCV[11], 23'AAAI workshop [12]

- LLM-Agent Design and Research.
  - Enhancing large models' capability to handle complex tasks through multi-round feedback and tool-calling mechanisms.
  - Publications: 25'NeurIPS[3], 25'MLCAD [7]

Graduate Assistant, Connecticut Transportation Institute, CT, United States **May. 2023 – May. 2024**

- Backend development for a large-scale roadway safety management system.

SKILLS

- **Programming language**: C/C++; Python; ASM; Verilog.
- **Software**: CUDA kernel design; PyTorch; Distributed Data Parallel; TensorFlow; MATLAB; Git; Compiling chain.

HONORS AND AWARDS

- **Predoctoral Fellowship**, UConn School of Computing (2025)
- **Eversource Fellowship**, UConn Eversource Energy Center (2024)
- **Cigna Fellowship**, UConn School of Engineering (2022)
- **1st Place in Accuracy & 4th Place Overall**, ACM/IEEE TinyML Design Contest (2022)
- **1st Prize**, Zhixin Cup National AI Robot Competition (2021)

PUBLICATIONS

1. [**25'ICLR**] **X. Xie**, Y. Luo, H. Peng, C. Ding. RTop-K: Ultra-Fast Row-Wise Top-K Selection for Neural Network Acceleration on GPUs. 2025 International Conference on Learning Representations.
2. [**25'ICCAD**] K. Thorat, H. Peng, Y. Luo, **X. Xie**, S. Huang, A. Hasan, J. Zhao, Y. Li, N. Wu, Z. Shi, C. Yu, C. Ding. GROOT: Graph Edge Re-growth and Partitioning for the Verification of Large Designs in Logic Synthesis. 2025 IEEE/ACM International Conference On Computer-Aided Design.
3. [**25'NeurIPS**] B. Lei, W. Kang, Z. Zhang, W. Chen, **X. Xie**, S. Zuo, M. Xie, A. Payani, M. Hong, Y. Yan, C. Ding. InfantAgent-Next: A Multimodal Generalist Agent for Automated Computer Interaction. 2025 Advances in Neural Information Processing Systems.
4. [**25'ICS**] Z. Li, H. Peng, X. Shen, M. Zabihi, **X. Xie**, G. Yuan, Y. Wang, O. Chen, C. Ding. Graph Convolutional Network Acceleration Using Adiabatic Superconductor Josephson Devices. 2025 ACM International Conference on Supercomputing.
5. [**25'ICS**] Y. Luo, S. Li, J. Tao, K. G. Thorat, **X. Xie**, H. Peng, N. Xu, C. Ding, S. Huang. DR-CircuitGNN: Training Acceleration of Heterogeneous Circuit Graph Neural Network on GPUs. 2025 ACM International Conference on Supercomputing.
6. [**25'WWW** RelWeb] C. Jin, H. Peng, A. Zhang, N. Chen, J. Zhao, **X. Xie**, K. Li, S. Feng, K. Zhong, C. Ding, D. N. Metaxas. RankFlow: A Multi-Role Collaborative Reranking Workflow Utilizing Large Language Models. Proceedings of the ACM on Web Conference 2025 (WWW-RelWeb 2025).
7. [**25'MLCAD**] K. Thorat, J. Zhao, Y. Liu, A. Hasan, H. Peng, **X. Xie**, B. Lei, C. Ding. LLM-VeriPPA: Power, Performance, and Area Optimization Aware Verilog Code Generation with Large Language Models. 2025 ACM/IEEE International Workshop on Machine Learning for CAD.
8. [**24'ASPLOS**] **X. Xie***, H. Peng*, K. Shivdikar, M. A. Hasan, J. Zhao, S. Huang, O. Khan, D. Kaeli, C. Ding. MaxK-GNN: Extremely Fast GPU Kernel Design for Accelerating Graph Neural Networks Training. 2024 ACM International Conference on Architectural Support for Programming Languages and Operating Systems.
9. [**24'ICCAD**] T. Zhou, J. Zhao, Y. Luo, **X. Xie**, W. Wen, C. Ding, X. Xu. AdaPI: Facilitating DNN Model Adaptivity for Efficient Private Inference in Edge Computing. 2024 IEEE/ACM International Conference on Computer-Aided Design.
10. [**23'ICCAD**] **X. Xie**, H. Peng, M. A. Hasan, S. Huang, J. Zhao, H. Fang, W. Zhang, T. Geng, O. Khan, C. Ding. Accel-GCN: High-Performance GPU Accelerator Design for Graph Convolution Networks. 2023 IEEE/ACM International Conference On Computer-

Aided Design.

11. [**23'ICCV**] H. Peng, S. Huang, T. Zhou, Y. Luo, C. Wang, Z. Wang, J. Zhao, **X. Xie**, A. Li, T. Geng, K. Mahmood, W. Wen, X. Xu, C. Ding. AutoReP: Automatic ReLU Replacement for Fast Private Network Inference. 2023 International Conference on Computer Vision.
12. [**23'AAAI** Workshop] H. Peng, S. Zhou, Y. Luo, N. Xu, S. Duan, R. Ran, J. Zhao, S. Huang, **X. Xie**, C. Wang, T. Geng, W. Wen, X. Xu, C. Ding. RRNet: Towards ReLU-Reduced Neural Network for Two-party Computation Based Private Inference. 2023 AAAI Workshop on DL-Hardware Co-Design for AI Acceleration.
13. [**Master's Thesis**] Use TensorFlow to implement an automatic phase picking method based on the nearest neighbor method, 2020.

PROFESSIONAL ACTIVITIES

- Reviewer for Conferences/ Journals

  Great Lakes Symposium on VLSI 2024 (Program Committee)
  ICLR 2025
  Alexandria Engineering Journal
  Journal of Organizational and End User Computing
  Jordanian Journal of Computers and Information Technology
  Journal of Systems Architecture
  Pattern Recognition
  Neurocomputing